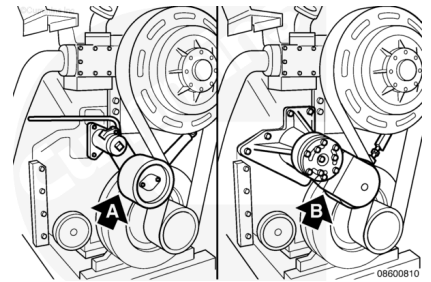


Trainings ▾

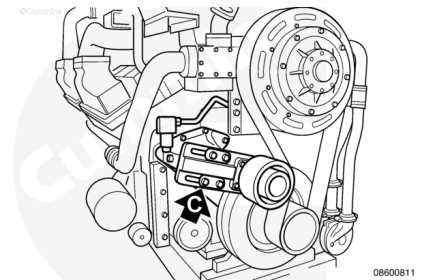
General Information

To assist in correct identification, there are three types of idler assemblies. A spring loaded type (A), a control arm type (B), and a slide arm type.



This illustration shows a slide arm type (C).

Some older engines may be retrofitted with the slide arm type of assembly.

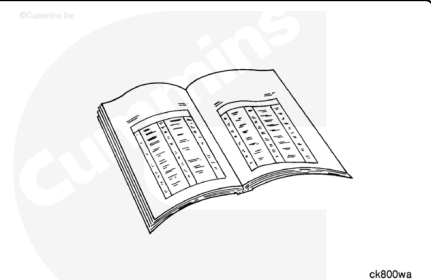


Preparatory Steps

⚠ WARNING ⚠

The fan belt idler is under tension. Do not allow hands to get between the idler and the belt or the fan hub. Personal injury can result.

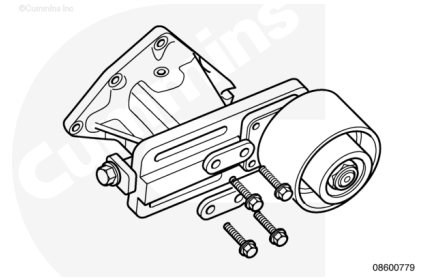
- Remove the fan. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html>)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html>)



Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

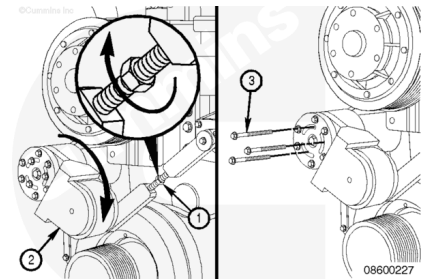


On engines fitted with the slide arm adjuster, remove the four capscrews from the adjusting bracket.

Remove the fan idler assembly and adjusting bracket, as an assembly.

⚠ WARNING ⚠

The belt tensioner and the pivot arm assembly will rotate during removal. Personal injury can result. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.

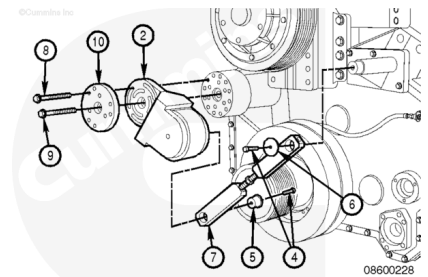


On engines fitted with the control arm adjuster, loosen the belt control rod (1) to release the idler arm assembly (2).

Remove the three capscrews (3) from the idler arm assembly (2).

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the two capscrews (4), retainer plug (5), washer (6), and the control rod (7).

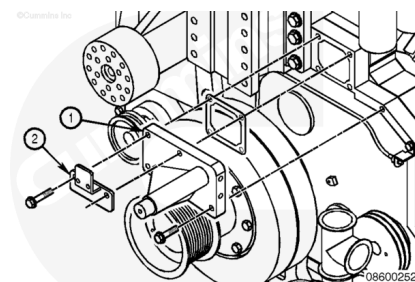
Remove the six capscrews (8).

Remove the capscrew (9), washer plate (10), and idler arm assembly (2).

For engines that use a Rockford™ fan clutch, the actuator bracket (1) has a solenoid bracket (2) attached by using the actuator bracket mounting capscrews. Both the solenoid bracket and actuator bracket are removed at the same time.

Remove the actuator bracket capscrews.

Remove the actuator bracket (1) and the solenoid bracket (2).



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

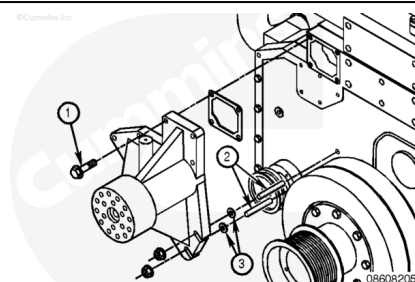
Use the correct lifting equipment to remove the idler pulley support bracket.

Remove the lower nuts from the idler pulley support bracket.

Remove the capscrews (1) from the idler pulley support bracket.

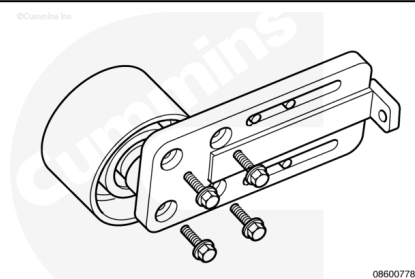
Remove the idler pulley support bracket.

Remove the spacers (3) and studs (2).



Disassemble

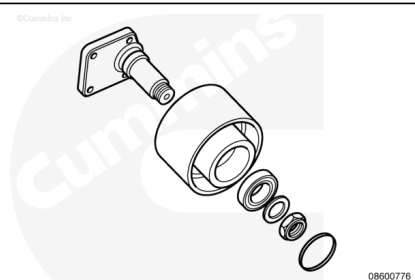
On engines fitted with the slide arm adjuster, remove the four capscrews from the rear of the adjusting bracket and detach the fan idler assembly.



Remove the locknut and washer from the shaft.

Remove the pulley from the shaft.

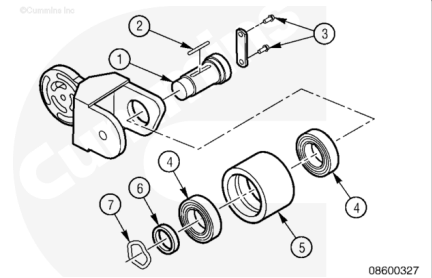
Remove the bearings, if necessary.



On engines fitted with the control arm adjuster, remove the two capscrews (3).

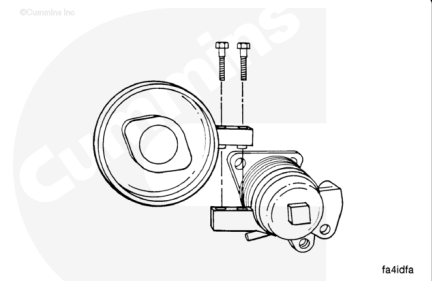
Remove the idler shaft (1), idler pulley (5), bearings (4), washer (6), and spring (7).

Remove the key (2) from the idler shaft (1), when present in the pivot arm assembly.



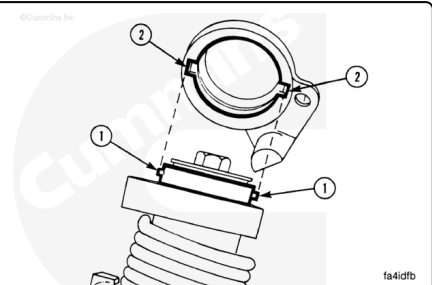
On engines fitted with the spring loaded adjuster, remove the capscrews.

Remove the pulley assembly from the pivot arm assembly.



Rotate the cap until the pins (1) align with the slots (2) in the cap.

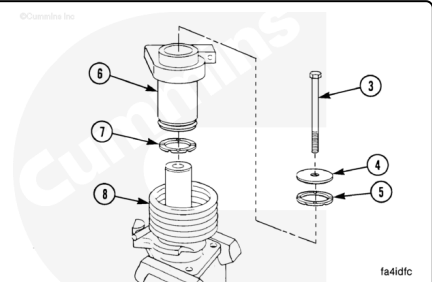
Remove the cap.



Use a vise to support the pivot arm assembly.

Remove the following parts:

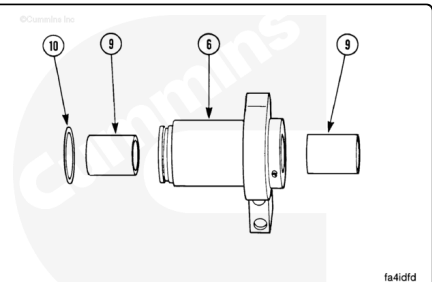
- Capscrew (3)
- Plain washer (4)
- Thrust bearing (5)
- Pivot arm (6)
- Thrust bearing (7)
- Spring (8).



Remove the following parts:

- Bushings (9)
- O-ring (10).

Discard the o-ring.

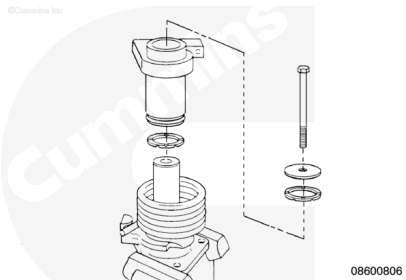


The pivot arm (6) contains an external wear sleeve to prevent the spring from contacting the arm. It can be replaced, if necessary.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



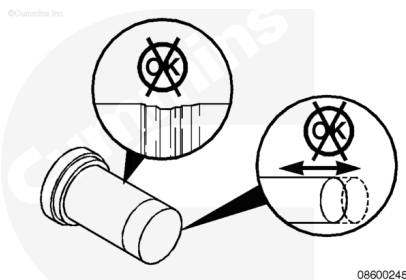
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

On engines fitted with the spring adjuster, clean the parts with solvent and dry with compressed air.

On engines fitted with the side arm type adjuster, check the shaft for wear or grooves on the surface. If the shaft has grooves in the surface or is loose, the shaft and the bearings **must** be replaced as an assembly.

Measure the shaft outside diameter.



Shaft Outer Diameter		
mm		in
49.96	MIN	1.967
49.98	MAX	1.968

If the idler was not disassembled, check the end clearance.

End Clearance		
mm		in
0.025	MIN	0.001
0.152	MAX	0.006

If the end clearance is **not** within specification, the idler **must** be rebuilt.

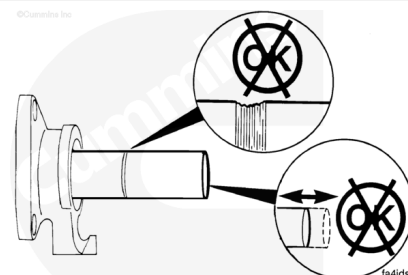
Rotate the idler to check for rough or damaged bearings.

If the idler does **not** rotate freely, the fan hub **must** be rebuilt.

On engines fitted with the spring-loaded adjuster, check the shaft for wear. Make sure the shaft is tight in the support.

If the shaft has grooves in the surface or is loose, the support and the shaft **must** be replaced as an assembly.

Measure the shaft outside diameter.



Fan Drive Idler Arm Assembly Outside Diameter

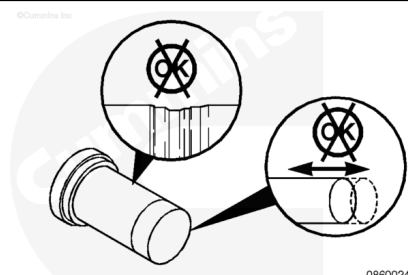
mm		in
38.087	MIN	1.500
38.113	MAX	1.501

On engines fitted with the control arm type adjuster, check the shaft for wear. Make certain that the shaft is tight in the support.

If the shaft has grooves in the surface or is loose, the support and the shaft **must** be replaced as an assembly.

Use a pivot arm assembly, when replacement is necessary.

Measure the shaft outside diameter.



Shaft Outer Diameter

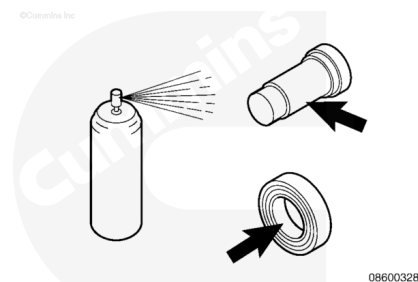
mm		in
59.97	MIN	2.361
57.99	MAX	2.283

Assemble

⚠ CAUTION ⚠

If disassembly is required after the Loctite™ has been applied and the bearings are locked onto the idler shaft, then the assembly must be heated to 177°C [350°F] to disassemble.

The bearings must be replaced after disassembly because this will exceed the approximate rated bearing temperature of 121°C [250°F].

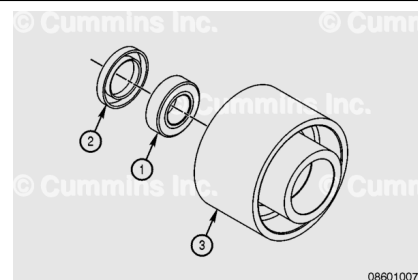


Use Loctite™ primer, Primer T - Loctite™, Part Number 3824715, to clean the idler shaft and the inside race of the bearings.

On engines with a slide arm adjuster, grease the bearing (1), if removed, before fitting. Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or equivalent.

If the above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

If removed, install the bearing (1) and oil seal (2) into the pulley (3).



Install the pulley (3) onto the shaft (4).

Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or equivalent.

If the above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Grease the cavity inside the pulley shaft with approximately 107 cc [4 oz] of grease. The grease level should fill approximately half way up the pulley shaft cavity.

Install the spacer (5) onto the shaft.

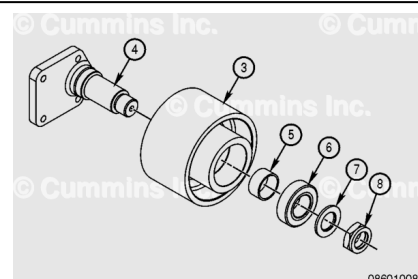
Grease the bearing (6) before fitting.

Install the bearing (7).

Install the washer (8) and locknut (9).

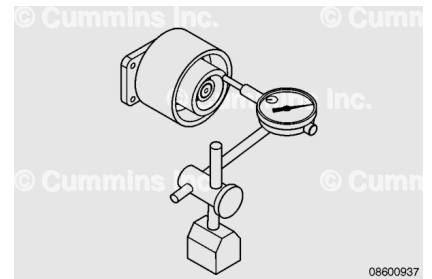
Tighten the locknut.

Torque Value: 542 n•m [400 ft-lb]



Use a dial indicator, Part Number 3376050, to measure the end clearance of the pulley.

Idler Arm Pulley End Clearance		
mm		in
0.025	MIN	0.001
0.152	MAX	0.006



If the end clearance is **not** within specification, remove the locknut, washer, second bearing inner race, and the spacer.

Install a spacer that will provide approximately 0.076 mm [0.003 in] of end clearance.

Install the bearing race, washer, and locknut.

Tighten the locknut.

Torque Value: 542 n·m [400 ft-lb]

Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

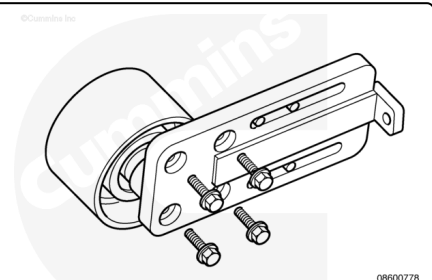
Grease the cavity at the end of the shaft with approximately 90 cc [3 oz] of grease. The grease level should fill approximately half way up the cavity (to the bottom of the locknut).

Install the metal end cover.



Use four capscrews to install the idler assembly to the adjusting bracket.

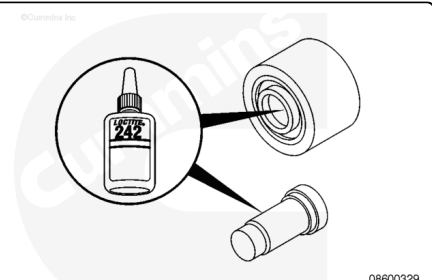
Torque Value: 260 n·m [191 ft-lb]



On engines with the spring-loaded adjuster, apply Loctite™ to the inside of the bearings and the idler shaft.

Use either of the two following bottle sizes of Loctite™ products:

- Part Number 3824041 - Threadlocker™ 242 (Blue) (Loctite™ 24221)



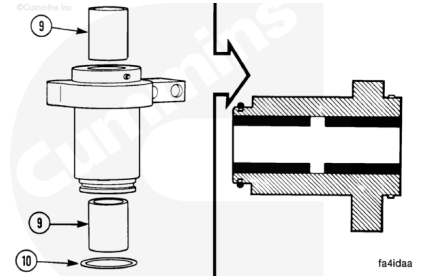
- Part Number 3824040 - Threadlocker™ 242 (Blue)
(Loctite™ 24231)

On engines fitted with the spring-loaded adjuster, it is recommended to use Loctite™ to secure the bearings to the shaft when reusing the pivot arm fan idler assembly.

Install and position the bushings (9) evenly into the pivot arm.

Lubricate the o-ring (10) with vegetable oil.

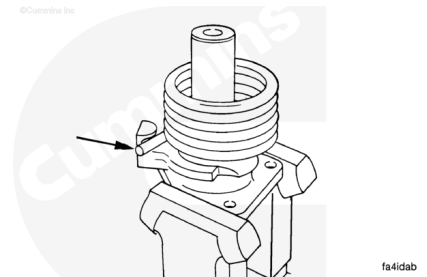
Install the o-ring onto the pivot arm.



Use a vise to support the pivot arm assembly.

Place the spring onto the support.

The end of the spring **must** be positioned on the spring stop on the pivot arm assembly, as shown in the illustration.



Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or equivalent.

If the above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

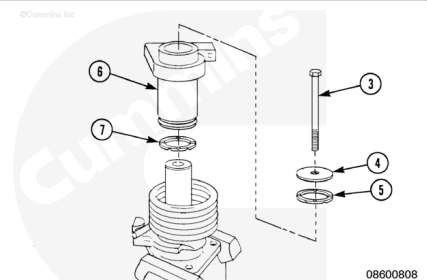
Lubricate the thrust bearings (5) and (7) with grease.

Install the parts:

- Thrust bearing (7)
- Pivot arm (6)
- Thrust bearing (5)
- Plain washer (4)
- Capscrew (3).

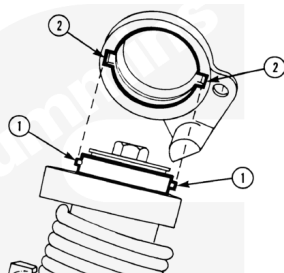
Tighten the capscrew.

Torque Value: 203 n•m [150 ft-lb]



Align the pins (1) with the slots (2) in the cap.

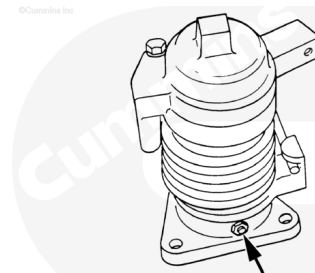
Install and rotate the cap.



fa4idfb

Lubricate the pivot arm assembly until grease begins to flow from under the cap.

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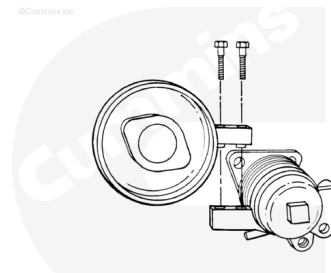
fa4idac

Install the pulley assembly onto the pivot arm assembly.

Tighten the capscrews.

Torque Value: 47 n•m [35 ft•lb]

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fa4idfa

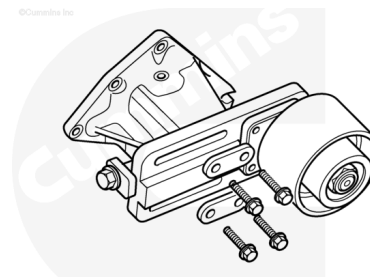
Install

On engines fitted with the slide arm adjuster, install the idler assembly and bracket on engine.

Install the four capscrews and two clamping plates.

Torque Value: 280 n•m [207 ft•lb]

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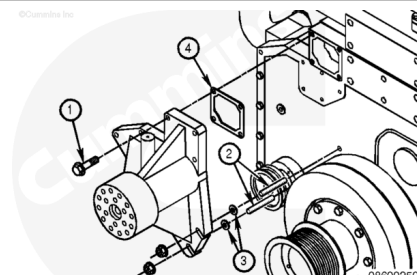


08600779

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.

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08600250

On engines fitted with the control arm adjuster, use the correct lifting equipment to install the idler pulley support bracket.

Make sure the rubber gasket (4) remains in the correct position during installation.

Install the studs (2) and spacers (3).

Install the idler support bracket and gasket (4).

Install the 12 mm capscrews (1) into the idler pulley support bracket and tighten.

Torque Value: 115 n•m [85 ft-lb]

Install the locknuts onto the studs (2) and tighten.

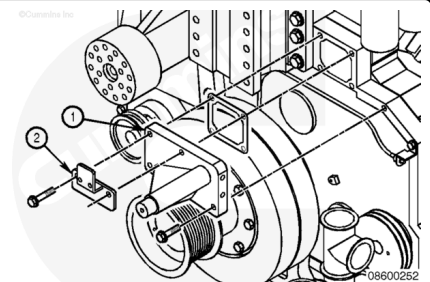
Torque Value: 80 n•m [59 ft-lb]

For engines that have a Rockford™ fan clutch, a solenoid bracket (2) is installed with the actuator bracket (1). The same mounting capscrews are used.

Install the actuator bracket capscrews, actuator bracket (1) and solenoid bracket (2) on the front gear cover.

Tighten the capscrews.

Torque Value: 115 n•m [85 ft-lb]



Position the idler arm assembly (2) with washer plate (10) and capscrews (9).

Install the six capscrews (8).

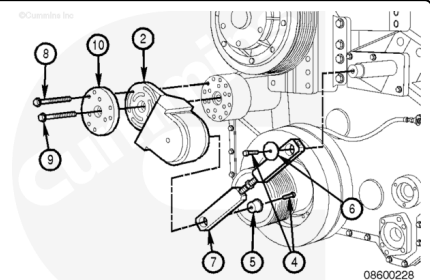
Tighten the capscrews.

Torque Value: 200 n•m [148 ft-lb]

Install the idler arm assembly (7), washer (6), retainer plug (5), and capscrews (4).

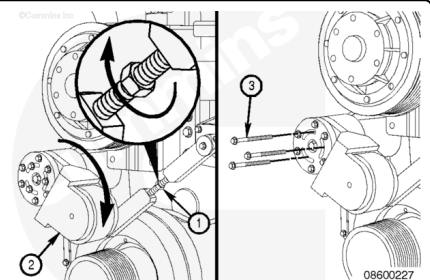
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Install the three capscrews (3) in the idler arm assembly (2).

Torque Value: 200 n•m [148 ft-lb]



Finishing Steps

- Install and adjust the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html>)
- Install the fan. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html>)
- Operate the engine and check for proper operation.

